

Bengali Product Review Sentiment Classification: A Classical Machine Learning Pipeline on a Self-Collected Dataset

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Abstract

This report presents an individually-built pipeline for classifying Bengali-language e-commerce product reviews as positive or negative sentiment. The dataset — 300 reviews, evenly balanced between positive and negative classes — was self-collected during independent research time and has not been published or peer-reviewed. Reviews were cleaned to retain Bengali script only, represented using TF-IDF features at the unigram, bigram, and trigram level, and classified using seven classical machine learning algorithms. Several models, most notably Multinomial Naive Bayes, achieve a perfect 100% accuracy, precision, recall, and F1 score across all three n-gram settings. We report this result transparently while treating it with appropriate caution: with only 30 examples in the test set and no cross-validation, a perfect score is more likely a sign of a small, vocabulary-polarized dataset than of genuine model robustness. We discuss this limitation explicitly and outline the dataset scaling and validation work that would be needed before drawing stronger conclusions.

1 Introduction

Sentiment classification for Bengali text remains comparatively under-resourced relative to English, with fewer large, publicly available, pre-labeled datasets. To explore this task independently, we collected 300 Bengali-language product reviews from an e-commerce platform, along with their associated star ratings, and manually derived binary sentiment labels. This report documents the full pipeline — data cleaning, feature extraction, model comparison — and, critically, provides a transparent discussion of a result that initially looks very strong but warrants careful interpretation.

This is an individual, self-directed project. The dataset has not been published, peer-reviewed, or used in any formal academic study; it should be read as an applied exploration rather than a validated research contribution.

2 Dataset

The dataset consists of 300 Bengali product reviews, perfectly balanced between 150 positive and 150 negative examples, each accompanied by a 1–5 star rating. Table 1 summarizes basic corpus statistics after cleaning.

Review length ranges from 4 to 62 words, with an average of 14 words. The most frequent words per class are strongly polarized: the negative class is dominated by *na* ("no/not", 78 occurrences)

Table 1: Dataset summary statistics (post-cleaning)

Class	Reviews	Total Words	Unique Words
Negative	150	2,050	817
Positive	150	2,058	617
Total	300	4,108	1,185 (combined unique)

and *baje* ("bad", 27 occurrences), while the positive class is dominated by *bhalo* ("good", 104 occurrences) and *dhonnobad* ("thanks", 66 occurrences). This vocabulary separation is revisited in Section 5, as it has direct bearing on how the classification results should be interpreted.

3 Methodology

3.1 Data Cleaning

Raw review text was cleaned using a regular expression retaining only Bengali Unicode script (range U+0980–U+09FF), removing Latin characters, punctuation, numerals, and emojis.

3.2 Feature Extraction

Cleaned text was vectorized using TF-IDF (term frequency–inverse document frequency), computed separately at three n-gram levels:

- **Unigram** (single words): 1,185 features
- **Bigram** (1- and 2-word sequences): 4,213 features
- **Trigram** (1-, 2-, and 3-word sequences): 7,572 features

3.3 Models

Seven classical machine learning classifiers were trained and compared at each n-gram level: Logistic Regression, Decision Tree, Random Forest (100 estimators), Multinomial Naive Bayes ($\alpha = 0.15$), K-Nearest Neighbors ($k = 3$), Linear SVM ($C = 0.2$), and RBF-kernel SVM ($C = 1000$, $\gamma = 0.00015$).

3.4 Evaluation Protocol

The dataset was split 90/10 (270 training, 30 test examples), with the same split used consistently across all seven models within each n-gram setting. Accuracy, Precision, Recall, and F1 Score were computed on the held-out test set.

4 Results

Tables 2, 3, and 4 report full results for the unigram, bigram, and trigram feature settings respectively. Bold values indicate a perfect 100% score.

Multinomial Naive Bayes achieves a perfect score across all three n-gram settings. Linear SVM reaches 100% under bigram features but collapses to 46.67% under trigram features — an unstable swing that is itself informative (Section 5).

Table 2: Performance — Unigram features

Model	Accuracy	Precision	Recall	F1
Logistic Regression	96.67	100.00	92.31	96.00
Decision Tree	93.33	92.31	92.31	92.31
Random Forest	96.67	100.00	92.31	96.00
Multinomial Naive Bayes	100.00	100.00	100.00	100.00
KNN	90.00	85.71	92.31	88.89
Linear SVM	90.00	100.00	76.92	86.96
RBF SVM	90.00	100.00	76.92	86.96

Table 3: Performance — Bigram features

Model	Accuracy	Precision	Recall	F1
Logistic Regression	96.67	100.00	92.31	96.00
Decision Tree	86.67	84.62	84.62	84.62
Random Forest	96.67	100.00	92.31	96.00
Multinomial Naive Bayes	100.00	100.00	100.00	100.00
KNN	90.00	91.67	84.62	88.00
Linear SVM	100.00	100.00	100.00	100.00
RBF SVM	90.00	100.00	76.92	86.96

5 Discussion

The headline result — several models reaching 100% on every metric — should not be read at face value as evidence of a production-ready classifier. We identify two specific reasons for caution, both grounded in the data itself rather than speculation:

Test set size. The held-out test set contains only 30 examples. At this scale, a perfect score carries substantial statistical uncertainty: a single misclassified review would already drop accuracy to 96.7%. A 30-example test set does not provide strong evidence of generalization.

Vocabulary separability. The most frequent words per class (Section 3, Table 1 discussion) show strong, almost non-overlapping polarization — *bhalo* ("good") and *dhonnobad* ("thanks") dominate positive reviews, while *na* ("no/not") and *baje* ("bad") dominate negative ones. A bag-of-words or n-gram model can exploit this separation very effectively on a small, clean dataset, without necessarily learning anything that would generalize to reviews using more neutral, mixed, or sarcastic language.

No cross-validation. All results come from a single train/test split. We cannot rule out that this particular split happened to be unusually easy; k-fold cross-validation would be needed to produce a more defensible estimate of true performance, and is a direct lesson carried over from a related project (BanglaBERT fine-tuning on a separate 400-example dataset), where a single-split evaluation was similarly found to be fragile.

The Linear SVM result under trigram features (46.67% accuracy, down from 100% under bigram) is a useful corroborating signal: it shows that performance on this dataset is sensitive to seemingly minor methodological choices, which is inconsistent with a model that has learned a robust, generalizable decision boundary.

Table 4: Performance — Trigram features

Model	Accuracy	Precision	Recall	F1
Logistic Regression	96.67	100.00	92.31	96.00
Decision Tree	86.67	90.91	76.92	83.33
Random Forest	96.67	100.00	92.31	96.00
Multinomial Naive Bayes	100.00	100.00	100.00	100.00
KNN	86.67	84.62	84.62	84.62
Linear SVM	46.67	44.83	100.00	61.90
RBF SVM	100.00	100.00	100.00	100.00

6 Limitations

- **Dataset size and scope.** At 300 reviews from a single e-commerce platform, the dataset is small and domain-specific; results should not be assumed to generalize to other review domains or platforms.
- **No cross-validation.** Results reflect a single 90/10 split; no variance estimate is available.
- **Self-collected, unpublished data.** The dataset has not been peer-reviewed or externally validated for labeling quality or collection methodology.
- **Class balance by construction.** The dataset was balanced by design (150/150); real-world review distributions are rarely this evenly split, so reported metrics may not reflect performance under realistic class imbalance.

7 Conclusion

This project builds a complete, working Bengali sentiment classification pipeline from a self-collected dataset, comparing seven classical ML models across three TF-IDF feature settings. While several models report perfect scores, we treat this as a signal to investigate rather than a result to showcase uncritically: the combination of a small test set, strongly polarized class vocabulary, and single-split evaluation means these scores most likely reflect the ease of this particular dataset rather than a validated, generalizable classifier. The most valuable next steps are methodological rather than architectural: scaling the dataset, introducing k-fold cross-validation, and testing on reviews with more ambiguous or mixed sentiment to obtain a realistic estimate of performance.